

PRANSHU RAJAN

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GitHub: github.com/pranshu-rajan | LinkedIn: linkedin.com/in/pranshu-rajan

SKILLS

Programming Languages: JavaScript, TypeScript, Python, SQL, HTML5, CSS3

Frontend Technologies: React.js, Next.js, Vite, Tailwind CSS, Framer Motion, GSAP, Three.js, React Three Fiber, Chart.js

Backend Technologies: Node.js, Express.js, REST APIs, JWT Authentication, OAuth2, NextAuth, Prisma ORM, Stripe, Razorpay, n8n Automation

Databases: PostgreSQL, Supabase, MongoDB, MySQL, Row Level Security, Database Triggers, Query Optimization

Cloud & DevOps: Vercel, Render, Railway, GoDaddy, Git, GitHub, CI/CD Pipelines

AI/Machine Learning: RAG Pipelines, Large Language Models (LLM), OpenAI, Groq, Claude API, Prompt Engineering, Vector Embeddings, pgvector, HNSW, BM25

Web Technologies & Tools: Leaflet, SEO Optimization, JSON-LD, Docker

Best Practices: Code Review, Unit Testing, Responsive Design, System Design, Team Leadership

EDUCATION

Electronics and Instrumentation Engineering | Nirma University, Gujarat

CGPA: 7.88 | Expected Graduation: July 2024 – Jul 2028

XII (GSEB) | Swastik Academy, Gujarat

Percentage: 80.3% | Year: 2024

PROFESSIONAL EXPERIENCE

Full Stack Developer Intern | Xtin Capital

May 2026 - July 2026

- Architected and redesigned React frontends to match company visual identity and improve user experience across fintech applications
- Engineered scalable Node.js and Express backend services with optimized database management and clean code architecture
- Implemented SEBI-compliant hybrid payment architecture utilizing Razorpay and MFU APIs for SIP mandate processing
- Resolved critical production bugs including data-integrity and race-condition issues affecting payment workflows
- Conducted responsive design audits across multiple devices and prepared formal QA test reports

AI & Cloud Intern | Edunet Foundation - IBM SkillsBuild

September 2025 - October 2025

- Developed end-to-end AI models using Python with IBM Cloud and Watson Studio platform
- Executed complete machine learning pipeline including data collection, preprocessing, model training, and evaluation
- Deployed trained models on enterprise cloud infrastructure following production-grade best practices

PROJECTS

UPI Offline Mesh — Peer-to-Peer Offline Payment Prototype

Technologies: Java, Spring Boot, PostgreSQL, Next.js, TypeScript, REST APIs, Docker

GitHub: github.com/pranshu-rajan/upi-offline-mesh | Live: upi-offline-rho.vercel.app

- Built an offline payment system that solves zero-connectivity UPI failures by routing encrypted transactions peer-to-peer across nearby devices until a connected phone automatically forwards them to the banking backend.

- Secured data in transit using hybrid encryption (RSA-2048-OAEP + AES-256-GCM), preventing intermediate relay devices from reading sensitive details (amounts, PINs) or altering data without failing cryptographic validation.
- Eliminated double-spending and replay attacks using SHA-256 ciphertext idempotency hashing and database optimistic locking (@Version), validated through multi-threaded concurrency tests and a Next.js live simulation dashboard.

PacketLens AI - Real-Time Network Forensics & DPI Platform

Technologies: C++17, FastAPI, Python, Next.js, TypeScript

GitHub: github.com/pranshu-rajani/dpi-packet-analyser | Live: dpi-packet-analyser.vercel.app

- Built a multi-threaded C++17 packet engine to parse raw PCAP files, track connections, and extract TLS/HTTP headers, eliminating bottlenecks when processing large traffic dumps.
- Developed a scalable backend to persist network telemetry and apply dynamic firewall rules, automating the generation of sanitized PCAP files with blocked traffic removed.
- Created a dashboard with a Wireshark-style packet inspector, synchronized hex viewer, and streaming security copilot, allowing engineers to triage network threats directly in the browser.

Pranshu's AI - Production Vector Database and Hybrid RAG Engine

Technologies: C++17, Python, FastAPI, Next.js, TypeScript, Groq API, SQLite

GitHub: github.com/pranshu-rajani/pranshu-ai | Live: pranshu-ai.vercel.app

- Built a custom vector engine using C++17 & Python with HNSW and KD-Tree indexing, cutting query latency to sub-milliseconds and eliminating reliance on paid vector databases.
- Engineered a document ingestion pipeline using FastAPI with recursive text chunking and BM25/Dense hybrid search (RRF), preventing context fragmentation and fixing keyword-misses during retrieval.
- Developed a full-stack web dashboard using Next.js 15, TypeScript, and Docker with drag-and-drop file uploads, real-time graph inspection, and latency benchmarks, making index debugging and document management seamless.

POSITIONS OF RESPONSIBILITY

Executive Committee Board Member | International Society of Automation

December 2025 - December 2026

- Promoted to Executive Committee Board based on demonstrated leadership and technical contribution record
- Lead coordination of technical events, workshops, and competitions as part of Nirma University's annual tech fest

Member | International Society of Automation

May 2025 - July 2026

- Coordinated and managed technical events, workshops, and competitions supporting student learning and skill development

CERTIFICATIONS & CREDENTIALS

SAP Certified Generative AI Developer - SAP

AI Agents Course - Hugging Face

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional - Oracle

Get Started with Databricks for Machine Learning - SimpliLearn

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate - Oracle

Certificate of Scholar - Nirma University